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Swiggy Data Analysis Using Web Scraping

Team Members

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Agenda

To collect restaurant data from the Swiggy platform using web scraping techniques, perform data preprocessing and exploratory data analysis, and analyze key factors such as restaurant ratings, delivery time, distance, cuisine, and location to derive meaningful insights about food delivery patterns and service efficiency.

Problem Statement

Exploratory Analysis of Swiggy Restaurant Data to Understand Delivery Performance and Customer Satisfaction Factors.

Introduction

- Food delivery platforms allow customers to order meals from a wide variety of restaurants through online applications and receive them quickly at their location.
- This project analyzes restaurant data from the Swiggy platform to understand delivery performance trends, rating patterns, cuisine popularity, and location-based restaurant distribution.

Why this Project ?

- Food delivery platforms provide many restaurant options, but analyzing restaurant performance and delivery efficiency is challenging because restaurants vary in ratings, delivery time, cuisine, and location.
- This project uses web scraping and exploratory data analysis to study Swiggy restaurant data and identify patterns in delivery performance, restaurant ratings, and customer preferences.

Dataset Overview

- The dataset was collected using web scraping techniques from the Swiggy platform. I used Python libraries such as Requests and BeautifulSoup to extract restaurant information.
- The collected data includes details such as restaurant name, rating, cuisine type, delivery time, delivery distance, and location.

Terminologies:

- **Restaurant Rating:** Customer satisfaction score given to restaurants.
- **Delivery Time:** Estimated time required for food delivery.
- **Cuisine:** Type of food offered by the restaurant.
- **Distance:** Delivery distance between the restaurant and customer.
- **Location:** Area where the restaurant operates.

Dataset Overview

The dataset was collected using web scraping from the Swiggy food delivery platform. Python libraries such as Requests and BeautifulSoup were used to extract restaurant information from different locations. The collected data includes details such as restaurant name, rating, cuisine type, delivery time, delivery distance, and location, which were used for further exploratory data analysis.

Raw Data

```
df.head()
```

	Restaurant	Rating	Cuisine	Delivery_Time	Location	Distance_km	Restaurant_ID	Delivery_Category
0	Pizza Hut	4.5	Pizzas	28	Kachiguda	1.6	408177	Medium Delivery
1	Wow! Momo	4.3	Momos,Chinese,fastfood,Asian,Beverages	30	Himayath Nagar	2.7	360967	Medium Delivery
2	Barbeque Nation	3.9	North Indian,Barbecue,Kebabs,Biryani,Street Fo...	42	ABIDS	1.6	603116	Slow Delivery
3	Domino's Pizza	4.6	Pizzas,Italian,Pastas,Desserts	25	Kachiguda	3.0	28768	Fast Delivery
4	NIC Ice Creams	4.6	Ice Cream,Desserts	21	Narayanguda	1.8	108986	Fast Delivery

The data included information such as restaurant name, rating, cuisine type, delivery time, delivery distance, and location for restaurants available on the Swiggy platform..

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Data Cleaning

```
<class 'pandas.core.frame.DataFrame'>
Index: 553 entries, 0 to 671
Data columns (total 8 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Restaurant         553 non-null    object
 1   Rating             553 non-null    float64
 2   Cuisine            553 non-null    object
 3   Delivery_Time      553 non-null    int64
 4   Location           553 non-null    object
 5   Distance_km        552 non-null    float64
 6   Restaurant_ID      553 non-null    object
 7   Delivery_Category  545 non-null    category
dtypes: category(1), float64(2), int64(1), object(4)
memory usage: 35.2+ KB
```

After collecting the raw data through web scraping, we performed data cleaning using Pandas to prepare the dataset for analysis.

This included removing duplicate records, handling missing values, and converting relevant columns such as rating, delivery time, and distance into appropriate numeric formats.

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Feature Engineering

To analyze the dataset effectively, several derived features were created.

	Restaurant	Rating	Cuisine	Delivery_Time	Location	Distance_km	Restaurant_ID	Delivery_Category
0	Pizza Hut	4.5	Pizzas	28	Kachiguda	1.6	408177	Medium Delivery
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Feature Engineering

To analyze the dataset effectively, several derived features were created.
For example, we categorized delivery distances into **near, medium, and far ranges**, and calculated relationships between **delivery distance and delivery time** to evaluate delivery efficiency.
These derived features helped in better understanding restaurant performance and delivery patterns.

Univariate Analysis

- This histogram shows the distribution of delivery times across restaurants available on the Swiggy platform.
- Most restaurants have delivery times between 20 and 40 minutes, indicating that food delivery services generally operate within this time range.
- Only a small number of restaurants have very short or very long delivery times, suggesting that most deliveries are optimized around a moderate time window.

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Univariate Analysis

- This graph shows the distribution of delivery distances for restaurants.
- Most restaurants are located within a short delivery distance, indicating that food delivery services are optimized to serve nearby customers efficiently.

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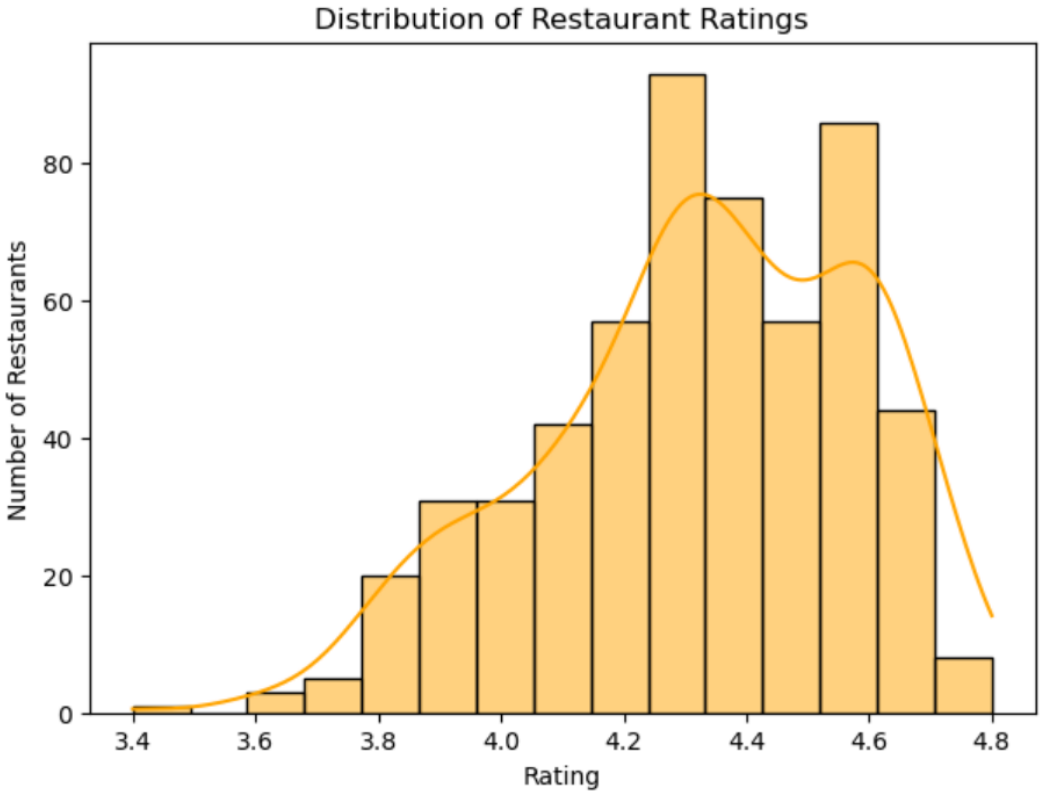
Univariate Analysis

- This histogram displays the distribution of restaurant ratings on the Swiggy platform.
- Most restaurants have ratings between 3.5 and 4.5, indicating that the majority of restaurants provide satisfactory service to customers.
- Only a few restaurants have very low or extremely high ratings, showing that most restaurants maintain moderate to good performance levels.



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Bivariate Analysis

- This scatter plot shows the relationship between delivery distance and delivery time.
- As the delivery distance increases, the delivery time also tends to increase. This indicates that distance is an important factor affecting delivery performance.

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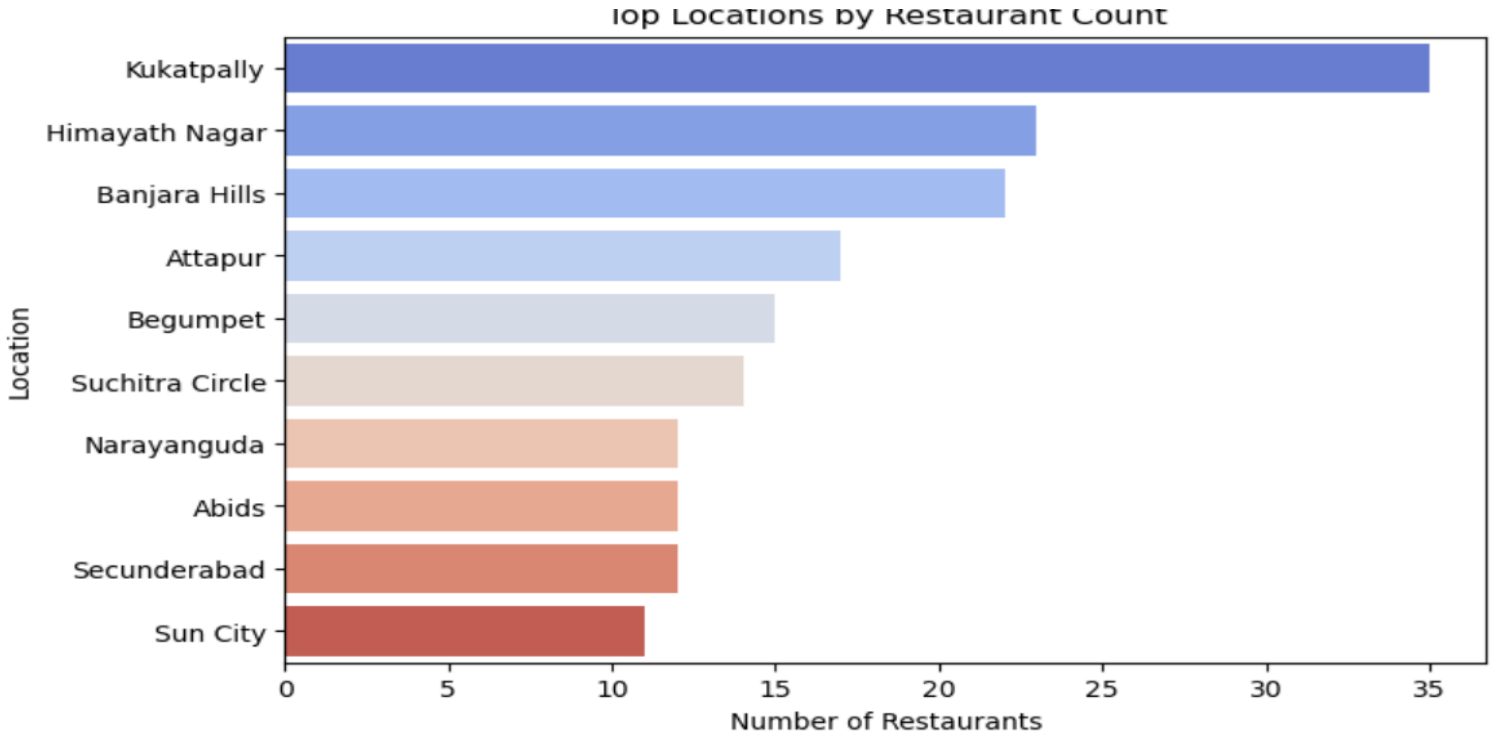
Bivariate Analysis



This chart shows the number of restaurants available in different locations on the Swiggy platform.

From the graph, we can observe that Kukatpally has the highest number of restaurants, followed by Himayath Nagar and Banjara Hills. These areas appear to have a higher concentration of restaurants, indicating strong food delivery demand.

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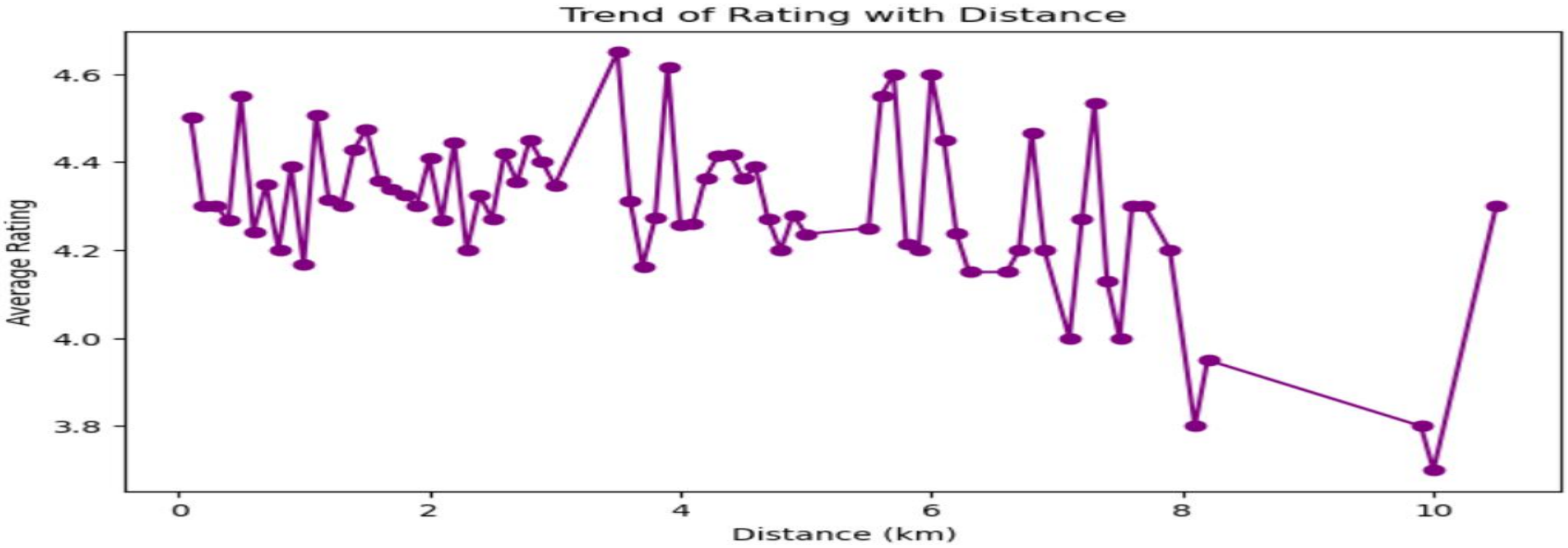
Bivariate Analysis



- From the graph, we observe that restaurants located at shorter delivery distances tend to maintain higher ratings, while ratings slightly decrease as the delivery distance increases.
- This suggests that longer delivery distances may affect customer satisfaction due to increased delivery time, which can influence restaurant ratings.

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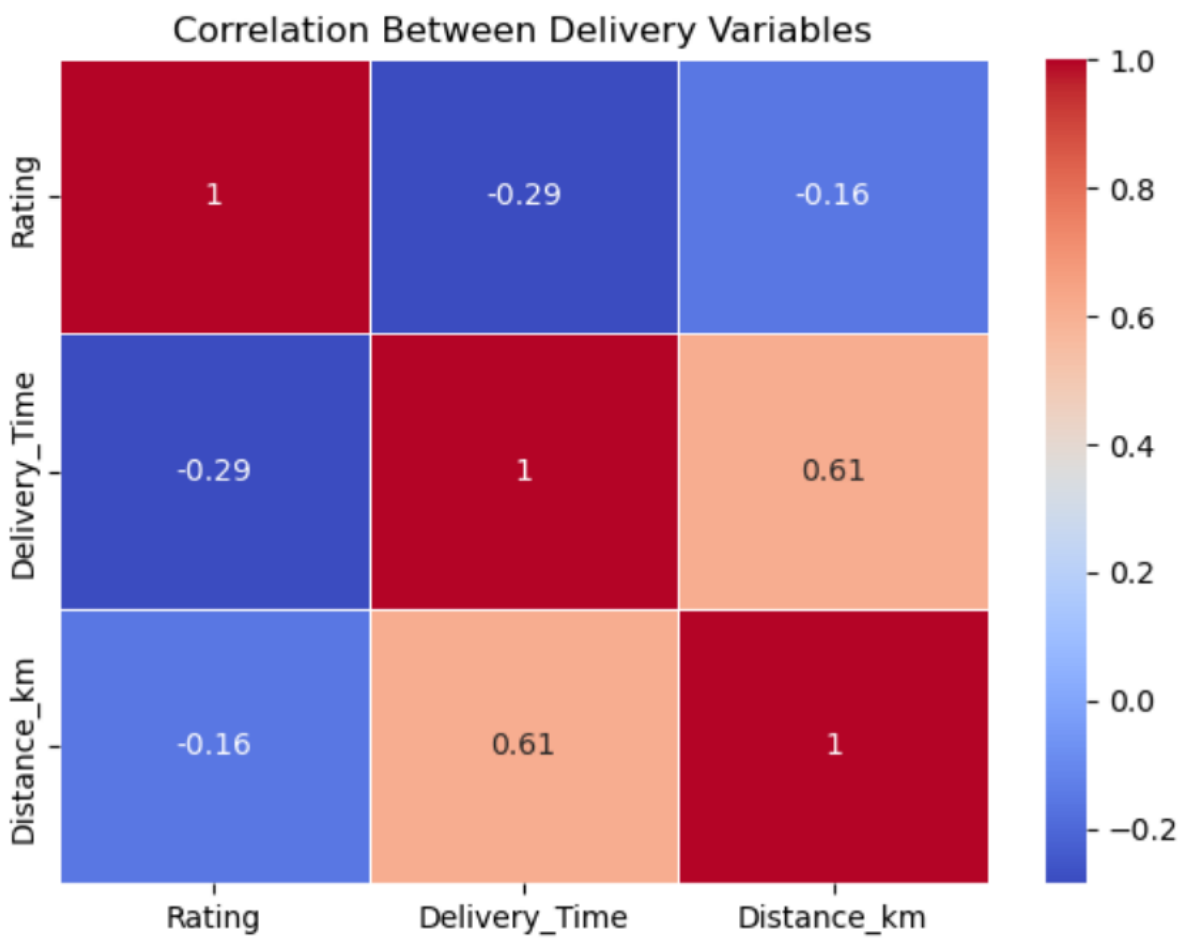
Multivariate Analysis



This heatmap shows the correlation between multiple variables such as rating, delivery time, and delivery distance.

From the heatmap, we can observe that delivery time is positively correlated with distance, meaning longer distances tend to increase delivery time.

The relationship between rating and delivery time appears relatively weak, indicating that ratings are influenced by multiple factors.

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Multivariate Analysis

- This scatter plot shows the relationship between delivery distance and delivery time, while the color indicates restaurant ratings.
- The graph shows that longer delivery distances generally increase delivery time, while restaurants with higher ratings are spread across different delivery distances.
- This visualization helps analyze how multiple factors together influence delivery performance.

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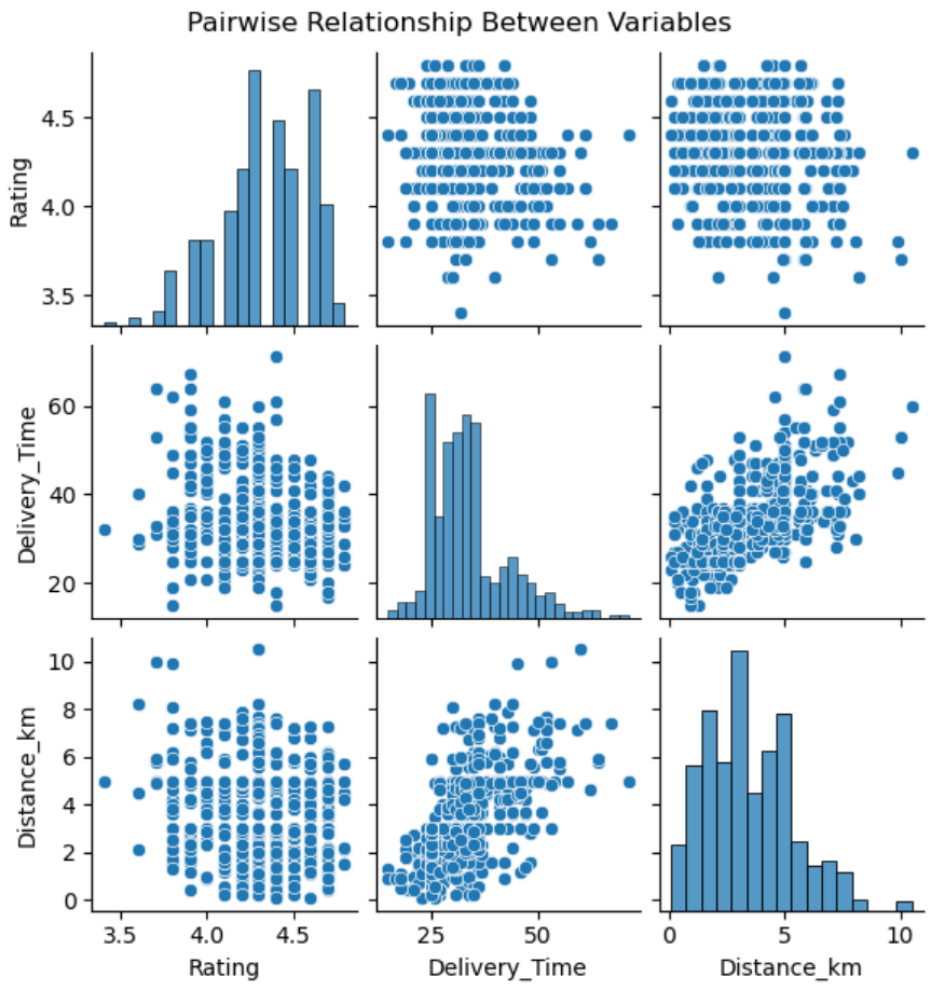
Multivariate Analysis



This pairplot visualizes the relationships between multiple variables such as rating, delivery time, and delivery distance simultaneously.
It helps identify patterns and trends between variables and provides a clearer understanding of how these factors interact in food delivery performance.

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Key Findings

- Most restaurants have ratings between 3.5 and 4.5, indicating that the majority of restaurants maintain good service quality.
- Delivery time generally ranges between 20 and 40 minutes, suggesting that most deliveries are optimized within this time window.
- Delivery distance has a noticeable impact on delivery time, as longer distances tend to increase the time required for food delivery.
- Certain locations such as Kukatpally, Himayath Nagar, and Banjara Hills show higher restaurant density, indicating strong food delivery demand in these areas.
- Popular cuisines dominate the restaurant listings, reflecting customer preferences and demand trends on the Swiggy platform.

Future Scope

- This project can be extended further by including additional factors such as customer order history, delivery partner performance, peak order timings, and customer feedback analysis.
- By incorporating these factors, we can develop predictive models to estimate delivery time, restaurant performance, and customer satisfaction. Advanced analytics and machine learning techniques could also be applied to improve delivery efficiency and demand forecasting.

Conclusion

- In conclusion, this project demonstrates how web scraping and data analysis can be used to study food delivery platforms.
- By analyzing restaurant ratings, delivery time, distance, and location distribution, we were able to identify patterns affecting delivery efficiency and restaurant performance.
- These insights can help understand customer behavior, restaurant popularity, and operational efficiency in food delivery services.

THANK YOU

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